SHRIPAD VILASRAO DESHMUKH

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(Last updated on Aug'23)

BACKGROUND

- Reinforcement Learning – *Decision Transformers, RL Interpretability, Agent-Based Models, Inverse RL*
- Computer Vision -- *Video Segmentation, Style Transfer, Document Analysis, Multi-Modal Element Association*

EDUCATION

University of Massachusetts Amherst, United States *Sept'23 -- Present*
M.S./Ph.D. in Computer Science

Indian Institute of Technology, Madras, Chennai, India *Aug'16 – Jul'20*
B.Tech. in Electrical Engineering, CGPA: *9.20/10*

CONFERENCE PUBLICATIONS

- S Singh, **S Deshmukh**, M Sarkar, B Krishnamurthy. "LOCATE: Self-supervised Object Discovery via Flow-guided Graph-cut and Bootstrapped Self-training". *Accepted at British Machine Vision Conference, 2023.*
- **S Deshmukh**, A Dasgupta, C Agarwal, N Jiang, B Krishnamurthy, G Theocharous, J Subramanian. "Explaining RL Decisions using Trajectories". *Published at International Conference on Learning Representations, 2023.*
- **S Deshmukh***, A Java*, M Aggarwal, S Jandial, M Sarkar, B Krishnamurthy. "One-Shot Doc Snippet Detection: Powering Search in Document Beyond Text". *Published at IEEE/CVF Winter Conference on Applications of Computer Vision, 2023.*
- P Gupta, N Puri, S Verma, D Kayastha, **S Deshmukh**, B Krishnamurthy, S Singh. "Explain Your Move: Understanding Agent Actions Using Focused Feature Saliency". *Published at International Conference on Learning Representations, 2020.*

WORKSHOP PUBLICATIONS

- **S Deshmukh**, Srivatsan R, S Vijay, J Subramanian, C Agarwal. "Counterfactual Explanation Policies in RL". *Published at ICML Workshop on Counterfactuals in Minds & Machines, 2023.*
- S Jandial, **S Deshmukh**, A Java, S Shahid, B Krishnamurthy. "Gatha: Relational Loss for enhancing text-based style transfer". *Published (**Oral**) at CVPR Workshop on Computer Vision for Fashion, Art, and Design, 2023.*
- S Singh, **S Deshmukh**, M Sarkar, R Jain, M Hemani, B Krishnamurthy. "FODVid: Flow-guided Object Discovery in Videos". *Published at CVPR Workshop on Learning with Limited Labelled Data for Image and Video Understanding, 2023.*
- **S Deshmukh**, A Dasgupta, C Agarwal, N Jiang, B Krishnamurthy, G Theocharous, J Subramanian. "Trajectory-based Explainability Framework for Offline RL". *Published at NeurIPS Offline RL Workshop, 2022.*

PATENTS

- S Singh, **S Deshmukh**, M Sarkar, B Krishnamurthy. "Novel Self-supervised Object Discovery in Videos". *Attorney Docket No. P12351-US, 2023 (Approved for filing)*
- S Ramesh, T Anand, **S Deshmukh**, J Subramanian, et al. "Account Executive Actionable Digest". *Attorney Docket No. P12314-US, 2023 (Approved for filing)*
- **S Deshmukh**, T Anand, J Subramanian, S Ramesh. "A Framework for Leveraging LLM Models and RL in Marketing Decision Making". *Attorney Docket No. P12223-US, 2023 (Approved for filing)*
- S Singh, **S Deshmukh**, M Sarkar, R Jain, M Hemani, B Krishnamurthy. "Video Object Segmentation through Flow-guided Graph-cut". *Attorney Docket No. P12170-US, 2023 (Filed)*
- S Singh, **S Deshmukh**, S Jandial, A Java, M Aggarwal, M Sarkar, A Jain, B Krishnamurthy. "FormProp: Novel method to propagate personalized error corrections across Forms corpus". *Attorney Docket No. P12004-US, 2022 (Filed)*
- **S Deshmukh**, A Dasgupta, C Agarwal, B Krishnamurthy, G Theocharous, J Subramanian. "Novel Trajectory-based Explainability Framework for RL-based Decision Making". *Attorney Docket No. P11853-US, 2022 (Filed)*
- A Java, **S Deshmukh**, S Jandial, M Aggarwal, M Sarkar, A Jain, B Krishnamurthy. "Forms Similarity Matching Framework for Enhancing RnC tool in AEM Forms" *Attorney Docket No. P11882-US, 2022 (Filed)*
- **S Deshmukh**, A Java, M Aggarwal, S Jandial, M Sarkar, A Jain, B Krishnamurthy. "A Novel Multimodal One-Shot Detection Approach for Document Snippet Search". *Attorney Docket No. P11686-US, 2022 (Filed)*

- S Jandial, **S Deshmukh**, B Krishnamurthy. "SelfDisturb: Novel Semantic Noise based Soft Label Regularization for Distilling Model Knowledge". *Attorney Docket No. P11539-US, 2022 (Filed)*
- **S Deshmukh**, S Jandial, A Java, M Aggarwal, M Sarkar, A Jain, B Krishnamurthy. "AdaData: Novel method to simplify data points for easier understanding of neural networks". *Attorney Docket No. P11364-US, 2022. (Filed)*
- S Jandial, S Ramesh, **S Deshmukh**, B Krishnamurthy. "Novel Method and Apparatus to Control Diffusion Model Image Generation". *Attorney Docket No. P11343 -US, 2022. (Filed)*
- **S Deshmukh**, M Aggarwal, M Sarkar, H Gupta. "Refining Element Associations for Form Structure Extraction". *Attorney Docket No. P10768-US, 2021. (Filed)*
- **S Deshmukh**, S Verma, J Subramanian, P Gupta, N Puri. "Customer Journey Management Using Machine Learning". *Attorney Docket No. P10405-US, 2021. (Filed)*

WORK EXPERIENCE

Machine Learning Research Associate, Adobe, Noida ***Feb'22– Jul'23***

Counterfactual Explanation Policies for RL. *In collaboration with C Agarwal* *Jan'23– May'23*

- In this project, we focused on using counterfactual explanations for RL policies. Basically, we aim to answer "what minimal change to my current policy would lead me to a certain desired outcome?" motivated by counterfactuals in XAI
- Proposed a complete mathematical framework, COUNTERPOL, by extending counterfactuals to RL, and presented a practical online Monte Carlo RL algorithm to achieve the target returns which can flexibly work in diverse state and action spaces
- **Established theoretical connection** between optimization for counterfactual explanation policy and trust region-based RL optimization. This result allows us to use RL optimization techniques such as TRPO, PPO, etc. for XAI purposes, and also helps us look at the prominent class of RL algorithms as a counterfactual estimation.
- Work is **published at ICML workshop** on counterfactuals, and is also under review at a top-tier AI conference

Decision transformers through the lens of RL. *In collaboration with J Subramanian* *Feb'23– May'23*

- Our aim here was to investigate whether the decision transformer's autoregressive style of sequence modelling is sufficient for RL problems, or if adding traditional RL losses improves the performance.
- Systematically extended traditional losses for state, action, reward and return-to-go predictions such as Bellman loss and policy gradient loss to the decision transformers
- Experimentally, we found that **RL losses consistently improve transformer-based decision making**. The work is currently under submission at a top tier AI conference.

Unsupervised video object segmentation. *In collaboration with M Sarkar* *Jan'23– Jul'23*

- The aim of this project was to identify salient objects in a video in a *fully* unsupervised fashion, i.e., without using any supervised labels in any possible way.
- We developed an approach where we utilised unsupervised object flow to guide a graph-cut on patch features of a frame. I ideated possible ways to build an end-to-end fully unsupervised segmentation pipeline.
- We built LOCATE, an approach that achieves state-of-the-art for fully unsupervised segmentation for videos as well as images on multiple benchmarks. Work has been **published at BMVC 2023.**

Text-based Style Transfer for Images. *Mar'23– Jul'23*

- In this project, we explored the idea of changing style of the image using text description. Main novelty with our approach is to relate the provided textual descriptions to a standard style template library for grounding.
- We proposed *Gatha* relational loss which has been very well received by the vision community with **Oral presentation awarded** at the prestigious CVPR workshop CVFAD. The work is also under submission at another top-tier CV conference.

Explaining RL Decisions using Trajectories. *In collaboration with G Theocharous & Prof. N Jiang (UIUC)* *Feb'22– Sept'22*

- Proposed a new explainability framework for RL -- finding past encountered trajectories that have led the agent to take certain action in a given state. Formulated a Shapley value-based framework to achieve this.
- Came up with a single representation for an RL trajectory using state-of-the-art sequence modeling (decision/trajectory transformer). Demonstrated utility of these embeddings in capturing semantically meaningful high-level behaviors.
- Backed up with strong results on *Grid-worlds*, *Atari* & *MuJoCo*, the work has been **published at ICLR 2023.**

Novel Snippet-based Document Search Paradigm. *In collaboration with M Sarkar* *Apr'22– Aug'22*

- To cater to the growing need for sophisticated document search utilities, we proposed a novel one-shot document snippet detection task – a task to identify snippets similar to queried document snippet
- Conceptualized visual similarity metric for documents that can programmatically generate huge query-target dataset required for training deep NNs. Metric helped resolve the issue of high subjectivity & cost involved in human annotations.
- Co-designed multi-modal cross-attention-based architecture, *MONOMER*, which beats state-of-the-art one-shot object detection & document analysis models (+ data generation heuristic in a real-world setting). **Accepted at WACV 2023.**

Agent-Based Models (ABMs) for Simulating Market Competition. *In collaboration with J Subramanian* *Mar'22– Jun'22*

- Implemented a simulation of the document-sign market to capture multi-agent interactions among the product competitors & consumers. Programmed the simulation from scratch, referring to ABM framework proposed in this paper.
- Researched particulars of product strategies & consumer behavior through consultancy reports. Simulation enabled us to systematically study emergent phenomena as market conditions are varied hitherto speculated using human intuition.

Machine Learning Engineer, Adobe, Noida *(Work from Home)* ***Jun'21 – Jan'22***

Shipment of 'List' and 'Section' detection models to Adobe's Automated Forms Conversion Service *Dec'21 – Jan'22*

- Deployed *Deformable DETR*-based list & section detection models leading to about 11% gain in class conversion quality

Conversion of Flamingo Forms Dataset to MS-COCO format *Sept'21 – Nov'21*

- Came up with an idea to convert *Flamingo Forms*, an Adobe proprietary document dataset, to standard *MS-COCO* format to be able to leverage the latest models of the fast-growing image recognition field
- Benchmarked performance of over 10 recent semantic segmentation + object detection approaches on 18 document classes. This exercise **boosted the average conversion quality of the entire service by 7%.**

Detecting Table Bounds in Form Documents *May'21 – Jul'21*

- Trained Semantic Segmentation models '*EfficientDet*', '*RetinaNet*', and Object Detection approaches '*YOLOv3*' & '*YOLOv5*' to capture bounds of tables within *blank* forms. With an improvement of over 9.5%, shipped the best model.

Member of Technical Staff, Adobe, Noida *(Work from Home)* *Aug'20 – May'21*

Risk minimization in Customer Journey Management using Inverse RL *Oct'20 – Dec'20*

- Ideated use of trajectory-based inverse RL methods for minimizing risk involved while intervening in a customer journey.
- Achieved more than 20% reduction in customer churn rate for batch RL settings**. Filed patent for the idea**.

Multimodal Element Association *Oct'20 – Feb'21*

- Debugged and deployed *TensorFlow 1.x* codebase for using segmentation masks as prior in a multi-modal element association pipeline. This resulted in 15% precision and 4% recall gain in extracting higher-order document structures.
- Drafted patent-idea for prior inclusion, '*Refining form structures…*' which is now part of **exemplary drafts** on *Adobe Brightidea* (the only new addition to the list in 2021).

INTERNSHIPS

Summer Research Intern, Adobe, Noida *May'19 – Aug'19*

Explaining Reinforcement Learning Agents. *Mentors: N Puri, S Verma, P Gupta*

- Collaboratively proposed a new saliency metric for visual explanations of RL policy, significantly outperforming prior art in terms of specificity and relevance of the generated saliency maps.
- Co-Produced explanations for *Stockfish* chess engine. Extended the idea to *LeelaChess-0*, *Minigo* Go agent, and Atari games agents trained for *Pong*, *Breakout,* and *Space-Invaders*. The research is **published at ICLR 2020.**

Research & Development (R&D) Intern, Center for Development of Advanced Computing (CDAC), Pune *Dec'18 – Jan '19*

Matrix and Graph Computations in High-Performance Computing. *Mentors: Prof. N Karmarkar, Dr. VCV Rao*

- Part of National Supercomputing Misson, a project aimed at creating a supercomputer cluster to empower Indian R&D
- Developed FORTRAN libraries to test & extract the performance of Intel Linear Algebra Kernels for matrices of varying specs
- Successfully integrated all the existent modules developed for the project under one Unified Object-Oriented Interface, along with complete documentation. Additionally, I worked on algorithms for generating hyper-graphs.

SCHOLASTIC AWARDS & ACHIEVEMENTS

- National Prize Certificate holder for top students at IIT Madras (<7%) *2017*
- Secured All India Rank (AIR) 323 in JEE Advanced (top 0.16%) and AIR 491 in JEE Mains (top 0.03%) *2016*
- Selected for Kishore Vaigyanik Protsahan Yojana Fellowship with overall national rank of 621 *2015*
- Ranked among the top 1% in the National Chemistry Olympiad conducted by Homi Bhabha Institute for Science Edu. *2015*
- Recipient of National Talent Search Exam (NTSE) Scholarship (Stage II - AIR 516, Stage I – Maharashtra State Topper) *2012*

RELEVANT COURSEWORK

Reinforcement Learning	Dynamic Games: Theory & Applications	Pattern Recognition & Machine Learning
Convex Optimization	Linear Algebra Techniques for Data Analysis	Probability, Statistics & Stochastic Process

MENTORSHIP & VOLUNTEERING EXPERIENCE

Member of organizing committee, DX Coding Competition for at Adobe, Noida *Mar'23*

- Organized a grand ML challenge for Indian UG students where I was involved in creating the problem statement & evaluation mechanism. Challenge was aimed at spreading awareness about Adobe's DX cloud and eventually recruit talent.

Mentor at Adobe, Noida *Summers - 22, 23*

- Mentored **5** research interns working on various problems in the field of RL

Moderator at Liberty, Equality, and Fraternity (LEAF) Society, IIT Madras *Jan'20 – May'20*

- LEAF is a society of 20+ students established to instill a culture of healthy discussions on diverse socio-economic issues faced by the world, and India in particular. Joined as a member in August 2019.
- Got elected as a moderator for Jan-May'20 semester. Conducted 8 discussion sessions during my tenure. Sessions included discussion on topics such as agrarian distress, the COVID-19 situation, deep dive on human rights, etc.

Coordinator at Extra Mural Lectures (EML), IIT Madras *Jun'17 – May'18*

- EML is an official IIT Madras student body that organizes lectures on various topics: Arts, Technology, Culture & the like.
- As a part of EML, got an opportunity to organize lectures by luminaries such as Orkut Büyükkökten (Founder of Orkut), Nirmala Sitharaman (then Defence Minister), Dr. Prakash Amte (Magsaysay Award Winner), D. Subbarao (Ex. RBI Governor).
- I also organized a lecture at SRM university, outside the IITM, as part of the initiative to expand EML nationwide.